Delivery performance review

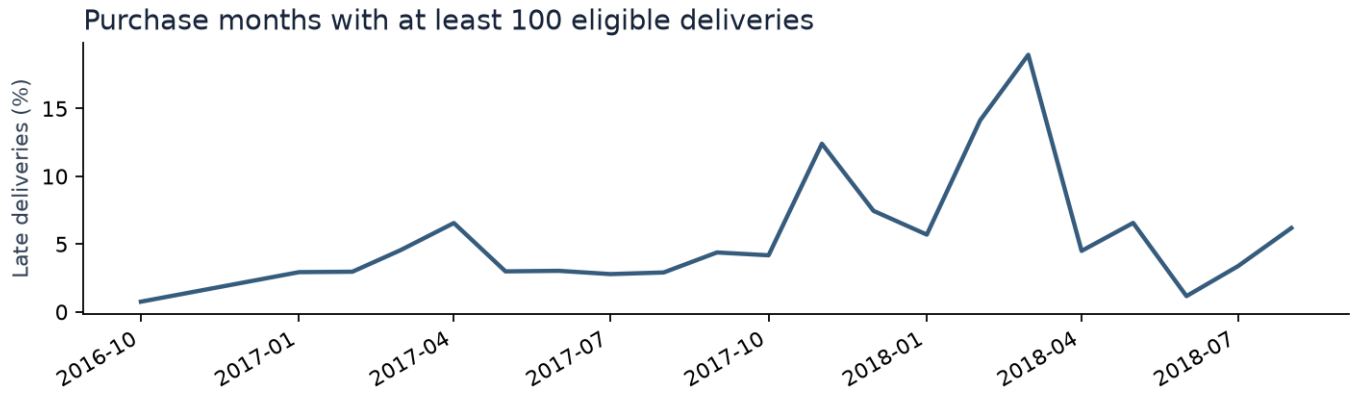
Predictive Delivery Risk, SLA Monitoring & Intervention Optimization

A decision-support prototype for an operations leader who needs to understand delivery reliability, spot risky purchases and allocate limited investigation capacity.

Source orders	On-time delivery	P90 delivery
99,441	**93.23%**	**23.1 days**

What the data establishes

The source contains 96,470 eligible completed deliveries, of which 6,534 arrived after the promised calendar date. Another 2,971 orders are reported separately because a valid completed-delivery outcome cannot be established.



The operating decision

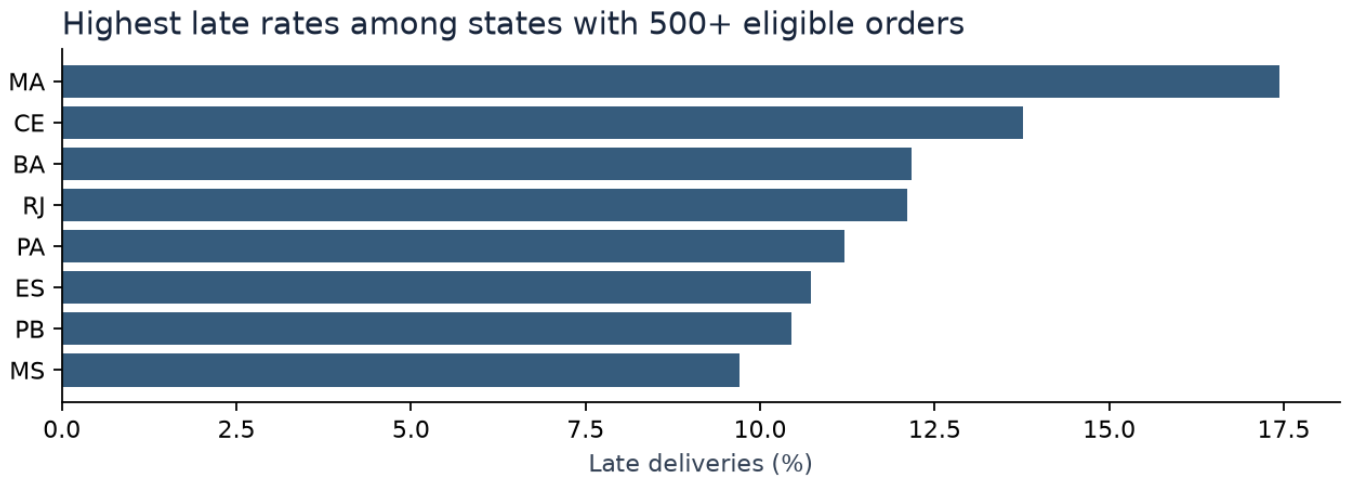
Use the dashboard to select an investigation cohort, review its sample size and risk, then allocate cases under an explicit daily budget. Validate proposed actions with a prospective intervention pilot.

Interpretation boundary

This is historical Olist data, not a live logistics feed. Item value, freight and payments are kept distinct. No actual intervention, avoided delay or realized cost saving is observed.

Regional and seller findings

Regional differences and seller performance are investigation signals. Order-level shipping milestones do not establish which seller or carrier caused a delay.



Protect the denominator and attribution

Seller comparisons use single-seller orders and publish sample sizes with Wilson intervals. Handoff/deadline diagnostics further restrict the cohort to one item and one seller. Primary-category reports assign each order to its first item category.

Data quality is visible

The warehouse reconciles order counts and independent item/payment totals. It flags 189 lifecycle anomalies rather than silently deleting them: 166 carrier timestamps precede purchase and 23 actual deliveries precede carrier handoff. Eight delivered-status orders lack actual delivery timestamps.

Recommended operational checks

1. Review high-rate regions with meaningful volume and compare primary-category strata.
2. Audit seller dispatch and carrier event recording before attributing accountability.
3. Investigate low review scores by delivery punctuality while retaining the reviewed-order denominator.

Uncertainty intervals describe observed rate uncertainty; they do not correct for all time/seller clustering or prove a causal regional effect.

Purchase-time model review

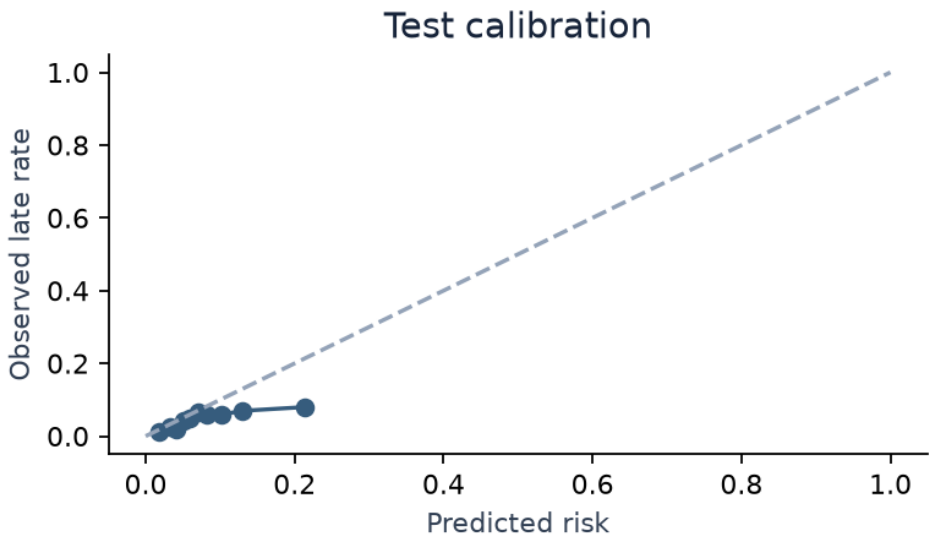
Checkout attributes and strictly earlier completed-delivery seller history enter the model. Actual delivery, handoff, duration, final status, payments and reviews are excluded from predictors.

Selected model	Test ROC-AUC	Test AP
Logistic	**0.634**	**0.072**

Chronology and label maturity

Training uses pre-January 2018 purchases. February fits calibration-method candidates; April forward validation selects the base/method pair; May defines final calibration and thresholds. Every fitting cutoff precedes July evaluation purchases. July-August evaluation is retrospective: those historical outcomes were examined in earlier versions.

The final test has 12,507 eligible deliveries and 4.81% late prevalence. Precision at 50 cases per purchase day is 5.69%, with recall 27.95%. The day-bootstrap 95% AP interval is 0.060-0.086.

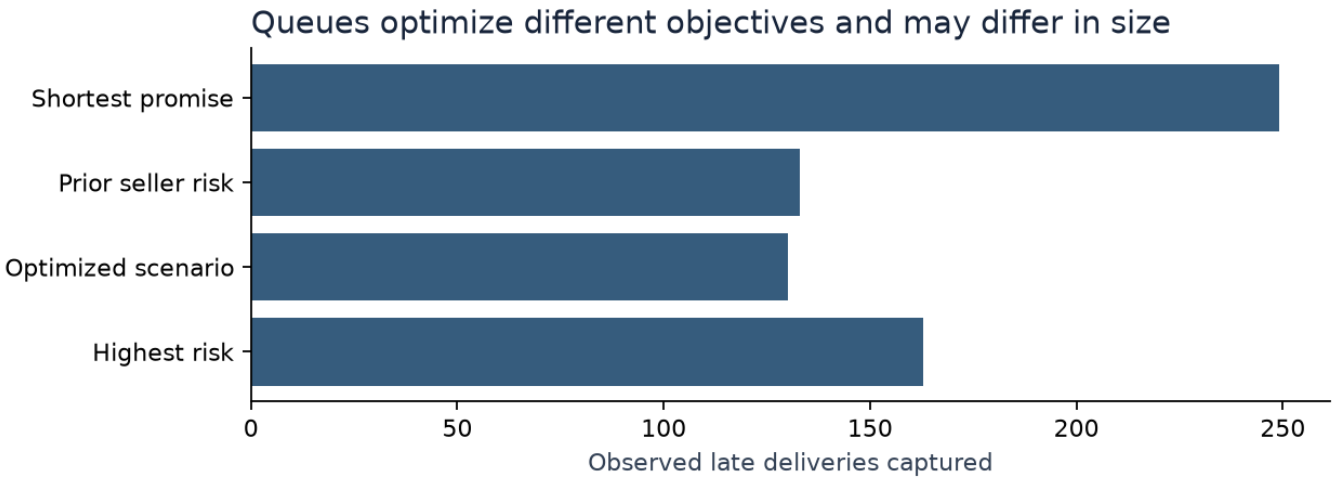


Revalidation completed but acceptance failed: selected probabilities overestimate the later cohort. Brier 0.048 exceeds the constant baseline 0.046. Ranking remains informative, but probability-based monetary scenarios require revalidation. Delivered-only labels and residual censoring also limit generalization.

Daily case planning

OR-Tools maximizes the sum of predicted risk x assumed effectiveness x assumed delay impact, minus investigation cost. Daily capacity, budget and regional caps constrain allocation; nonpositive value can leave capacity unused.

Default scenario: 50 cases/day, BRL 500/day budget, BRL 5/case, 20% effectiveness, and impact of BRL 150 plus 25% of item value. Sensitivity scenarios use 10% and 30% effectiveness. All these quantities are assumptions.



Observed capture versus simulated value

Highest-risk, shortest-promise, seller-history and seeded-random policies obey the same limits. Offline late-case capture is observed only where labels exist. Expected net benefit is a scenario estimate, not a measured saving. Different queue sizes prevent a simplistic precision-only comparison.

Delivered implementation and remaining acceptance

Included: PostgreSQL/dbt warehouse, Python pipeline, trained model, strict-time features, scenario allocator, six-page Streamlit dashboard, executed notebooks, tests, CSV/Parquet outputs, a portable report and an editable six-page Power BI project.

Native Desktop refreshed all seven tables and passed 6,101 DAX reconciliations. A required report-format file was fixed. Graphics capture was unavailable, so no pixel-level screenshot acceptance is claimed. Historical probability revalidation failed its preset gates; operational use remains unapproved.

Next step: validate checkout event immutability and operational costs, then run a controlled intervention pilot with calibration and customer-experience guardrails. Source: kaggle.com/datasets/olistbr/brazilian-ecommerce (Olist, 2018).